

RestroVibes — UML Process Model

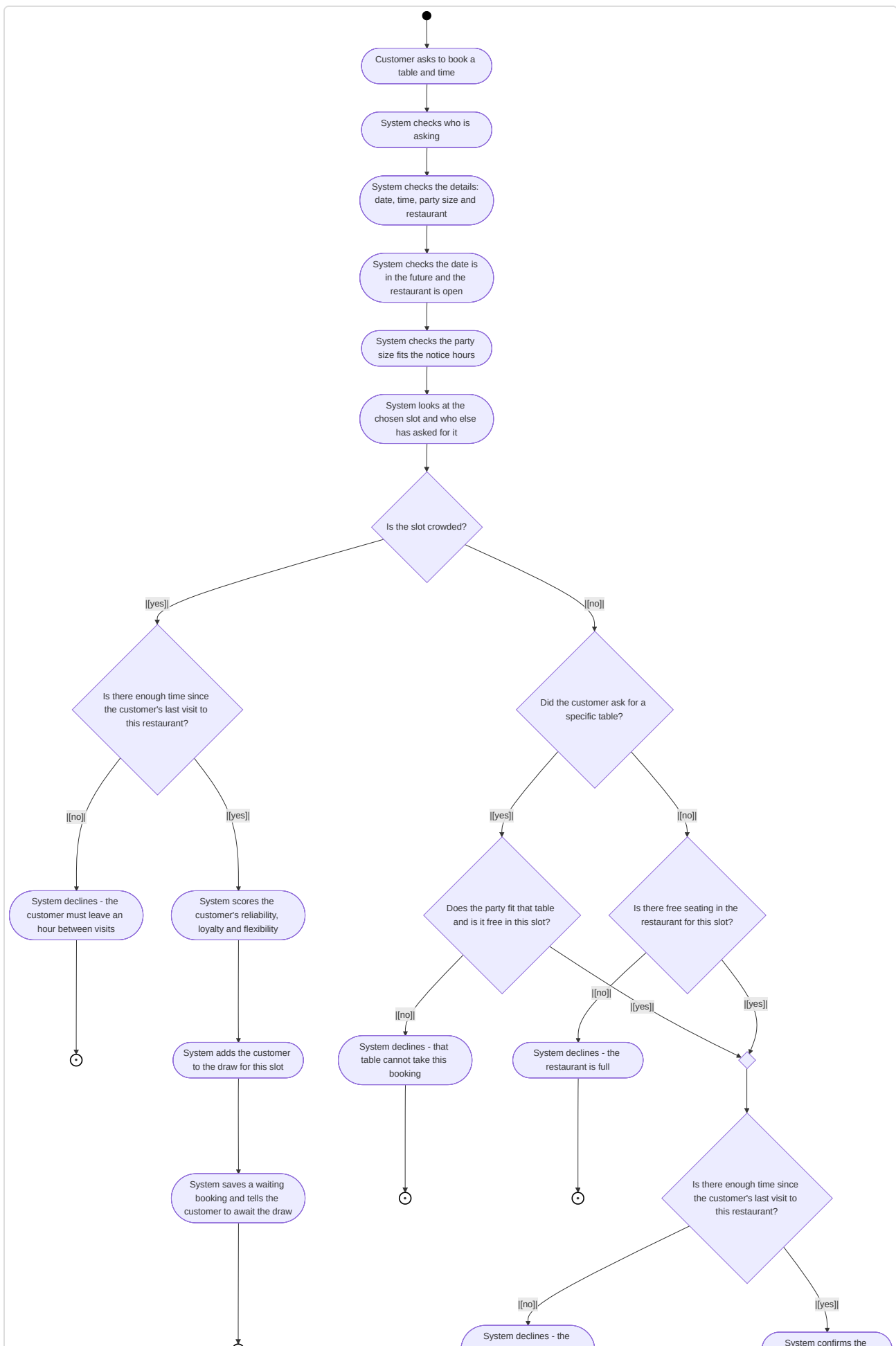
Plain-language UML 2.5 activity and state-machine models of how the platform works: booking a table, the draw for busy slots, what happens during and after a visit, cancellations, and how the platform keeps a trust score for each customer. The models describe *what happens*, not *how the code is written*.

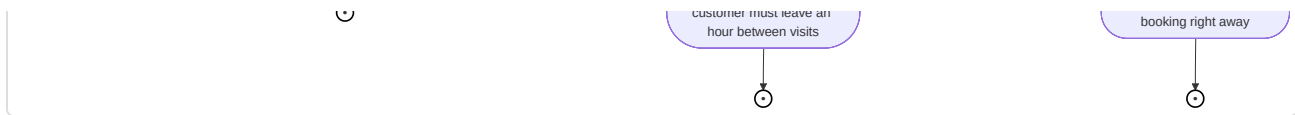
Notation Key

UML node	Notation	Meaning
Initial node	filled circle	where a process begins
Activity	rounded rectangle	a step or action
Decision / merge	diamond	a fork in the road (with guards) or joining of two paths
Guard	[condition] on an arrow	the question that decides which path to take
Final node	circle-in-circle	end of a process
Loop	backward arrow	the process repeats
State	rounded rectangle (state machine)	where a booking currently stands
Transition	what triggers it / what it does	how a booking moves from one state to another

The boxes also say who is acting: Customer, Restaurant, System.

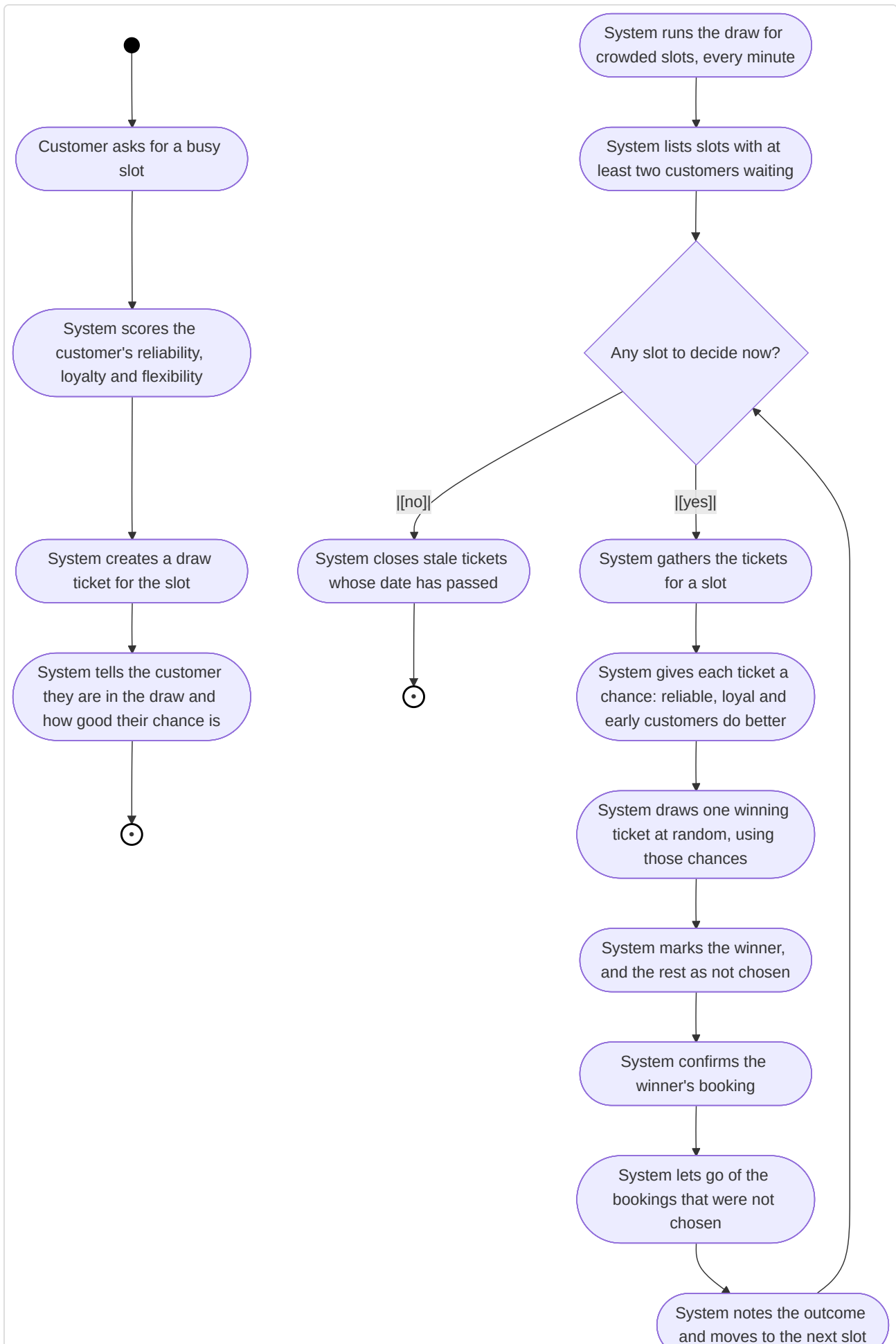
1. UML Activity Diagram — Booking a Table





- A slot is *crowded* when another booking or draw request already covers it, so more than one customer is competing for the same table and time.
- When the slot is crowded, the system puts the customer into the draw automatically. When it is free, the booking is confirmed immediately.

2. UML Activity Diagram — The Draw for a Busy Slot

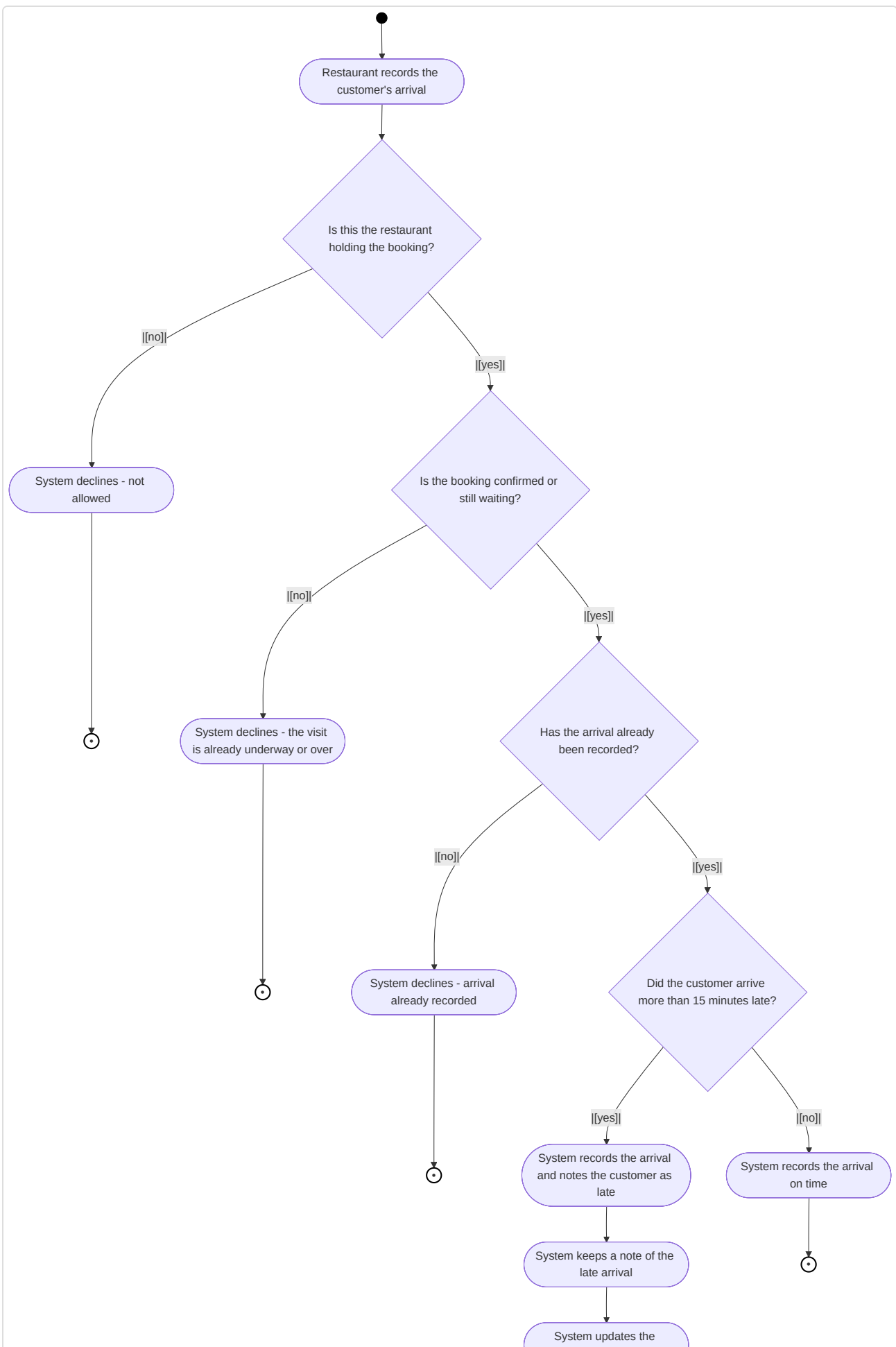


What makes a customer's chance better

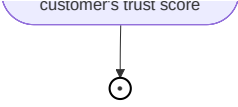
Factor	Effect
Everyone starts equal	base score of 100
Flexible about time, date or party size	higher score (+up to 50)
Regular, loyal customer	higher score (+up to 30)
History of missed, late or cancelled visits	lower score (up to -200)
Asked earlier than others	small extra advantage
Lowest possible score	chance never falls below 1

3. UML Activity Diagram — During and After the Visit

3.1 The Customer Arrives

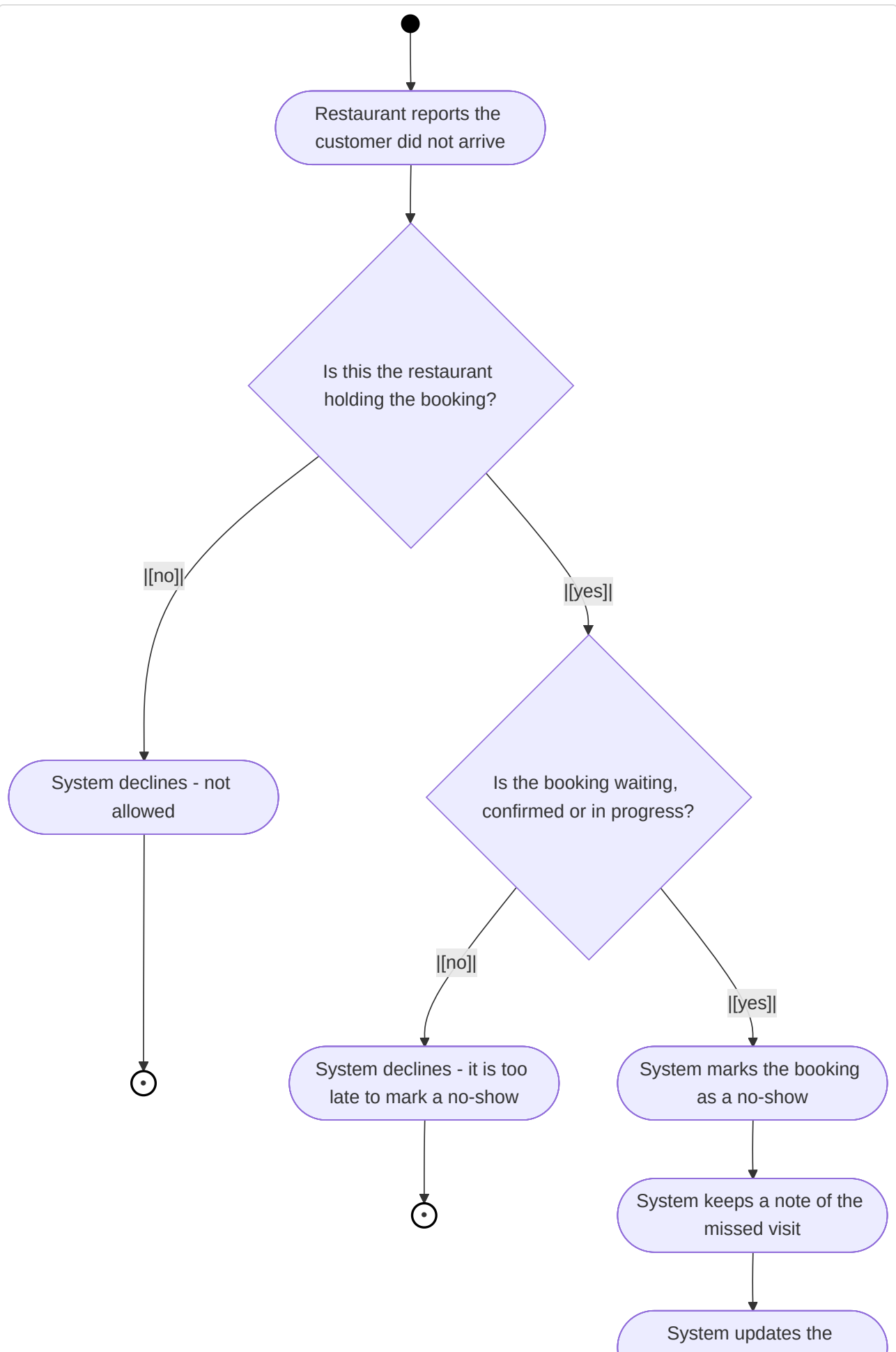


customer's trust score



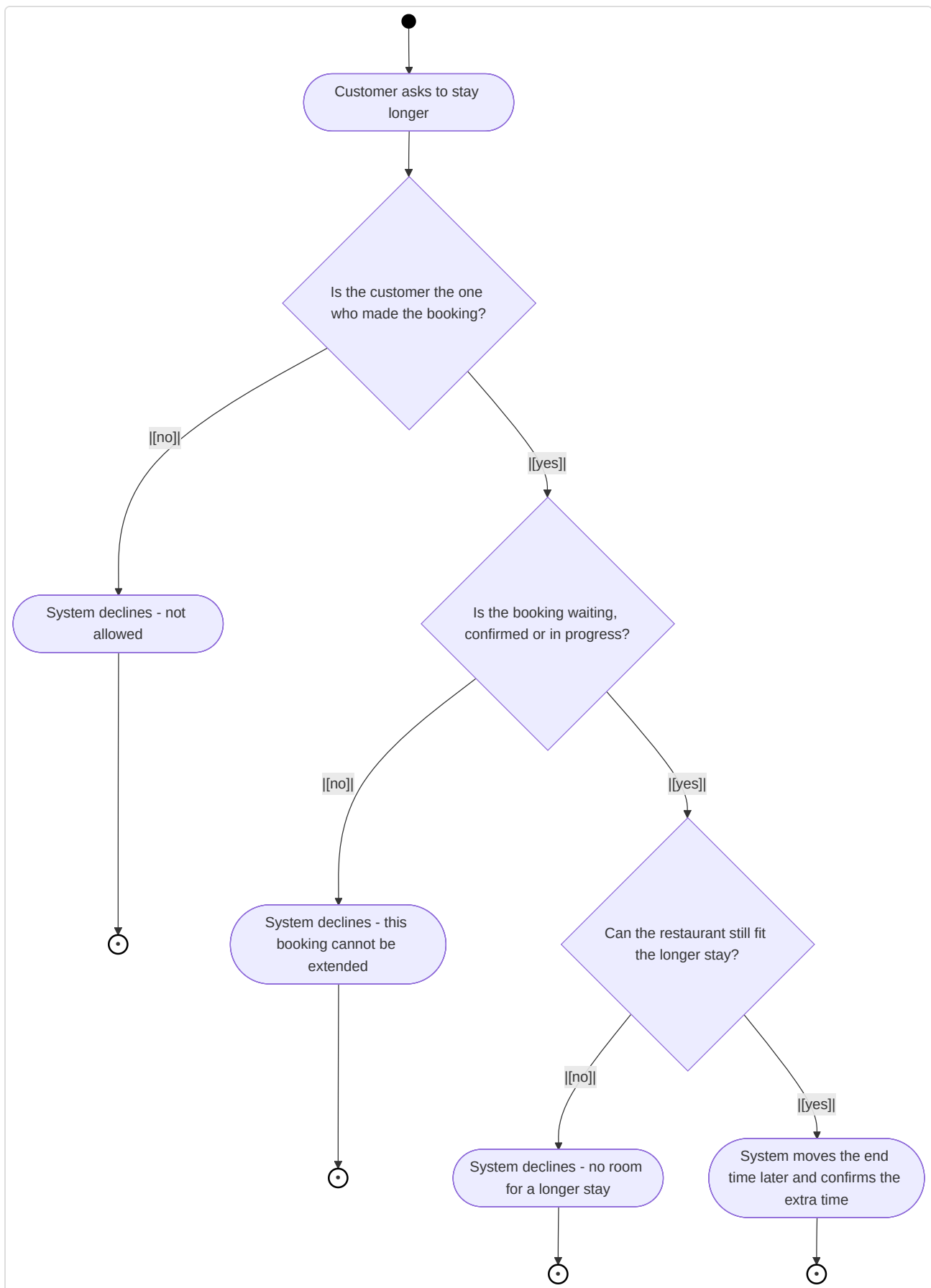
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graph TD; A(customer's trust score) --> B(( ));
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3.2 The Customer Does Not Arrive (No-Show)



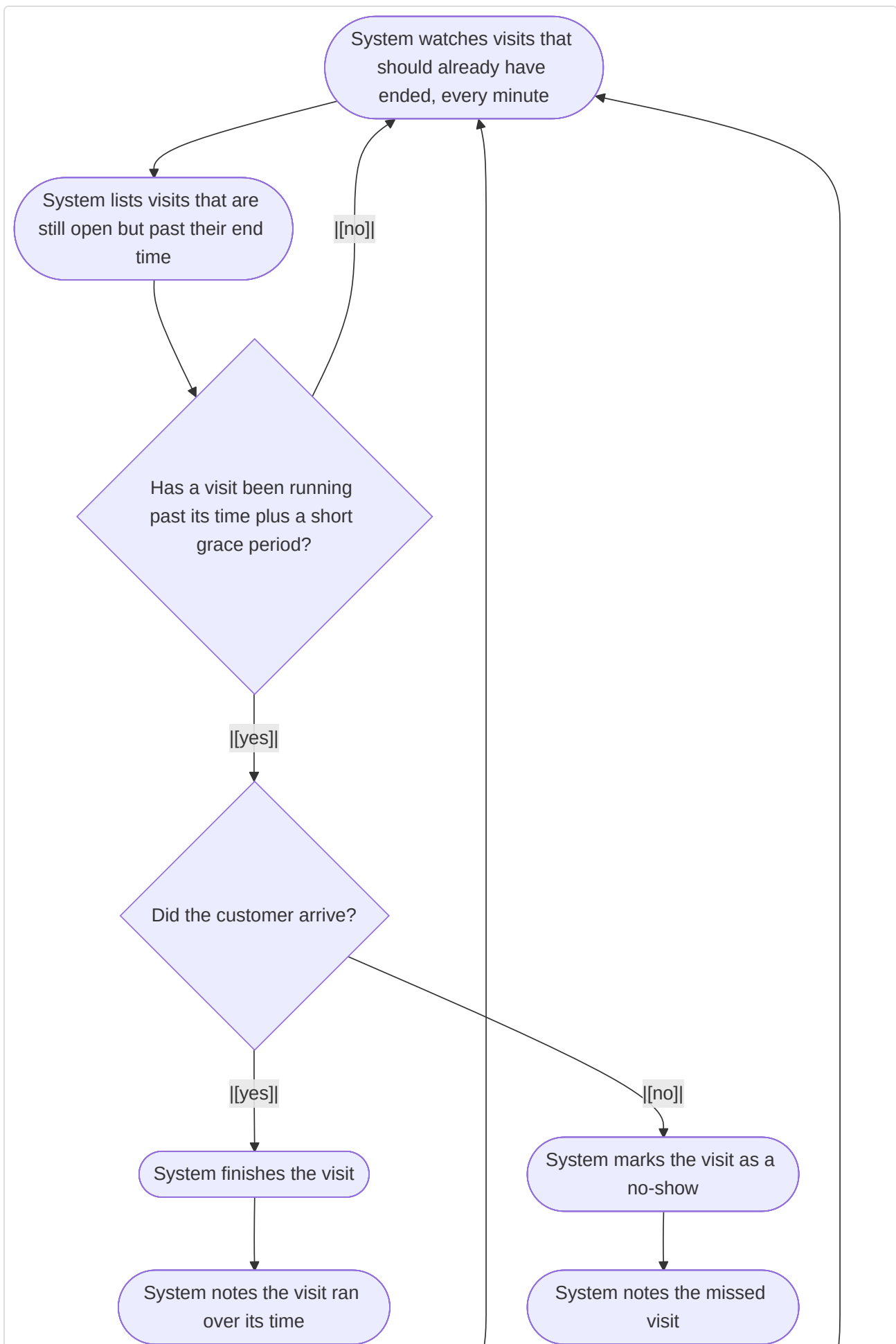


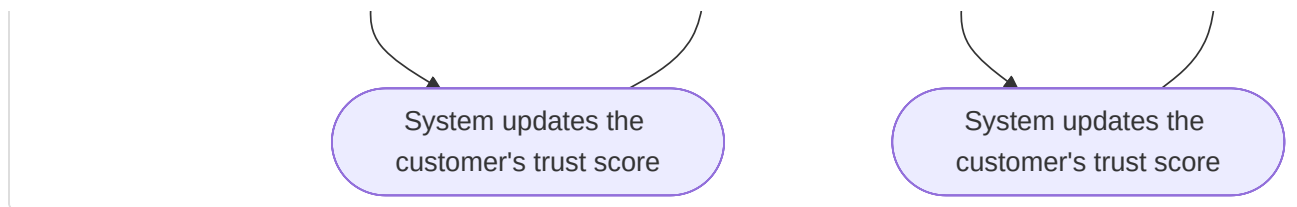
3.3 The Customer Stays Longer



3.4 The Visit Runs Past Its Time

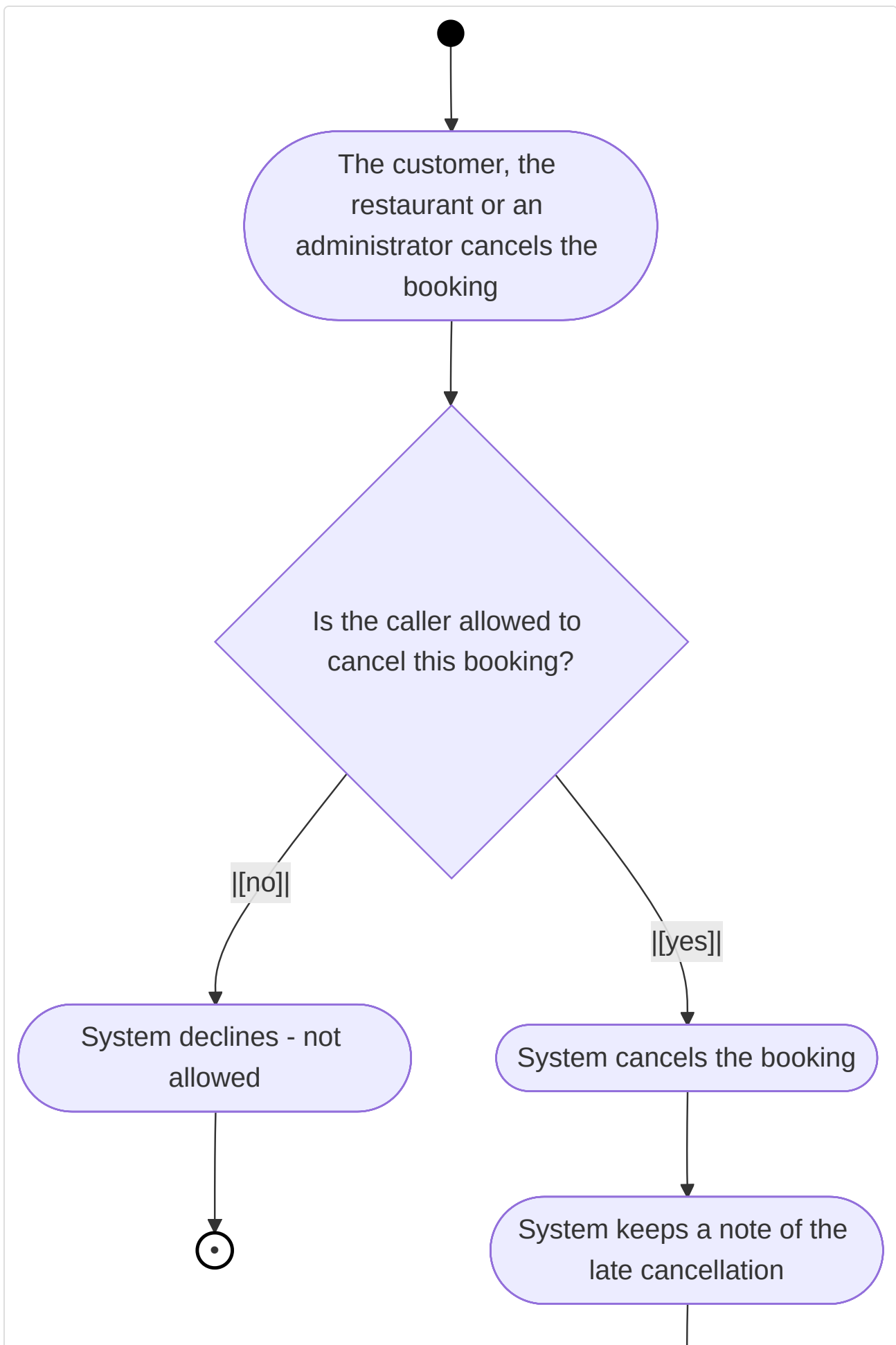
The system keeps watch in the background. It has no end point by design: each check finishes, then the next one starts a minute later.

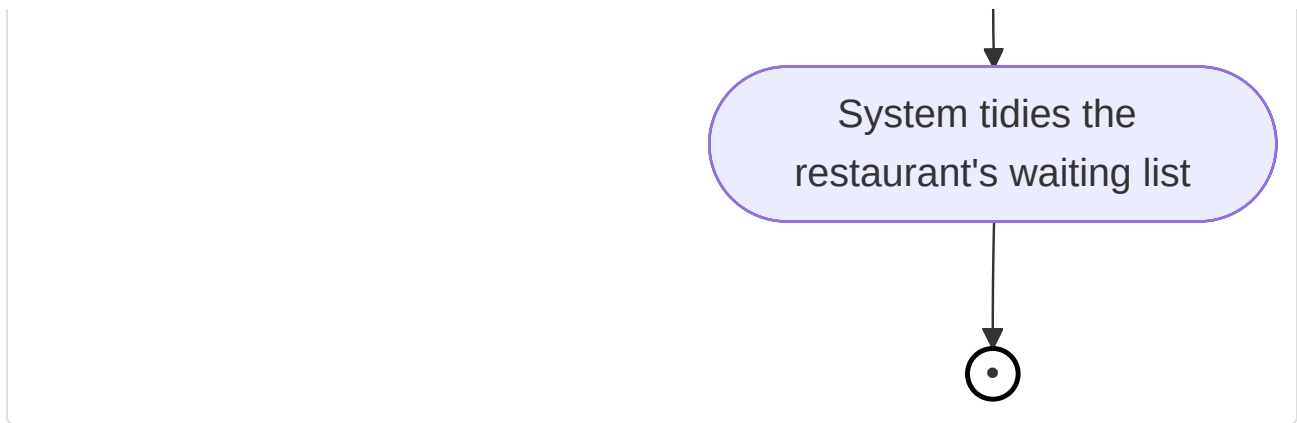




A restaurant can also finish a visit by hand at any time. In addition, a background job confirms the single waiting booking for a free slot after a short wait, while crowded slots are always handed to the draw.

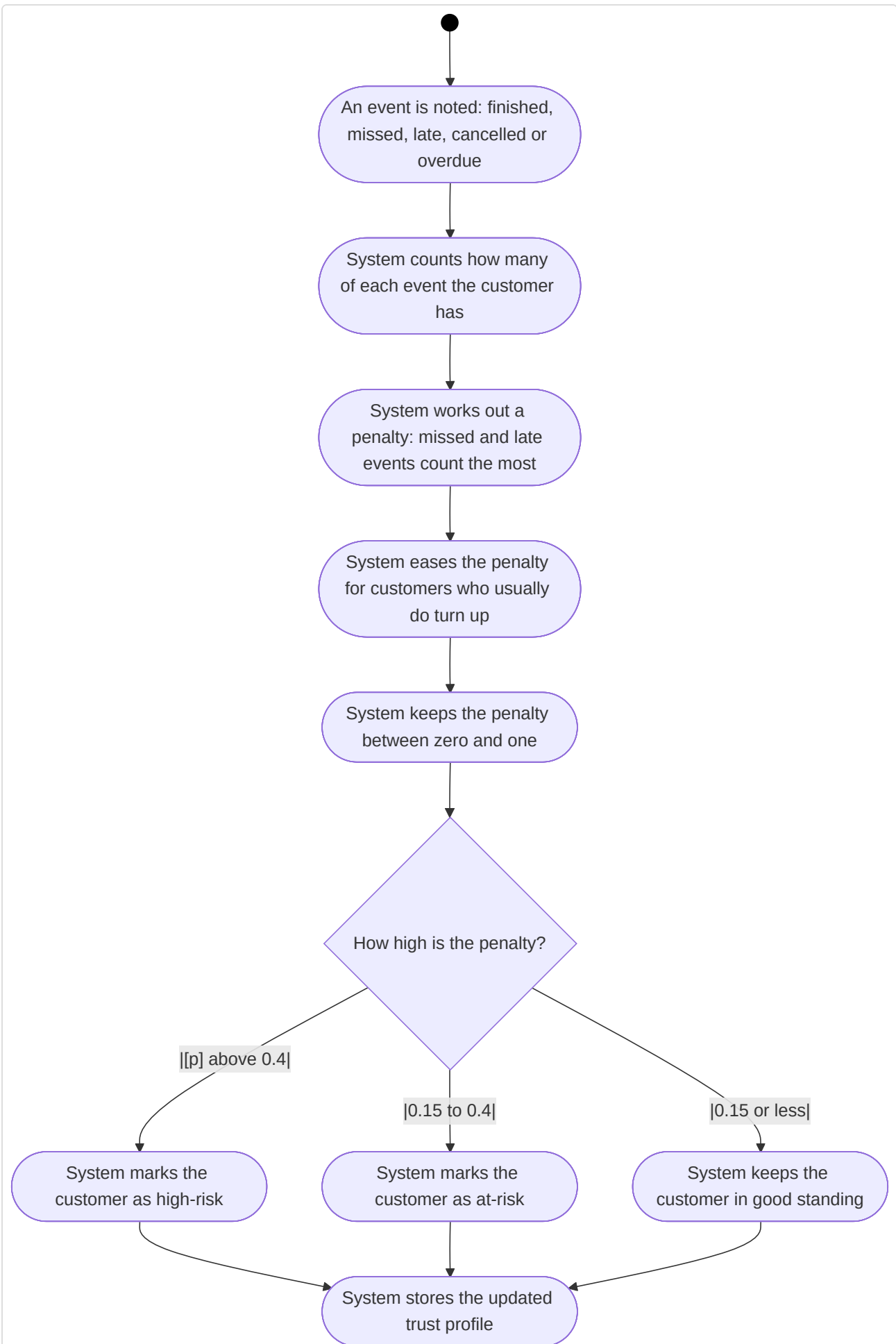
4. UML Activity Diagram — Cancelling a Booking





5. UML Activity Diagram — Keeping a Customer's Trust Score

Every visit that finishes, is missed, is late or is cancelled is noted. The system turns those notes into a simple trust score for the customer.

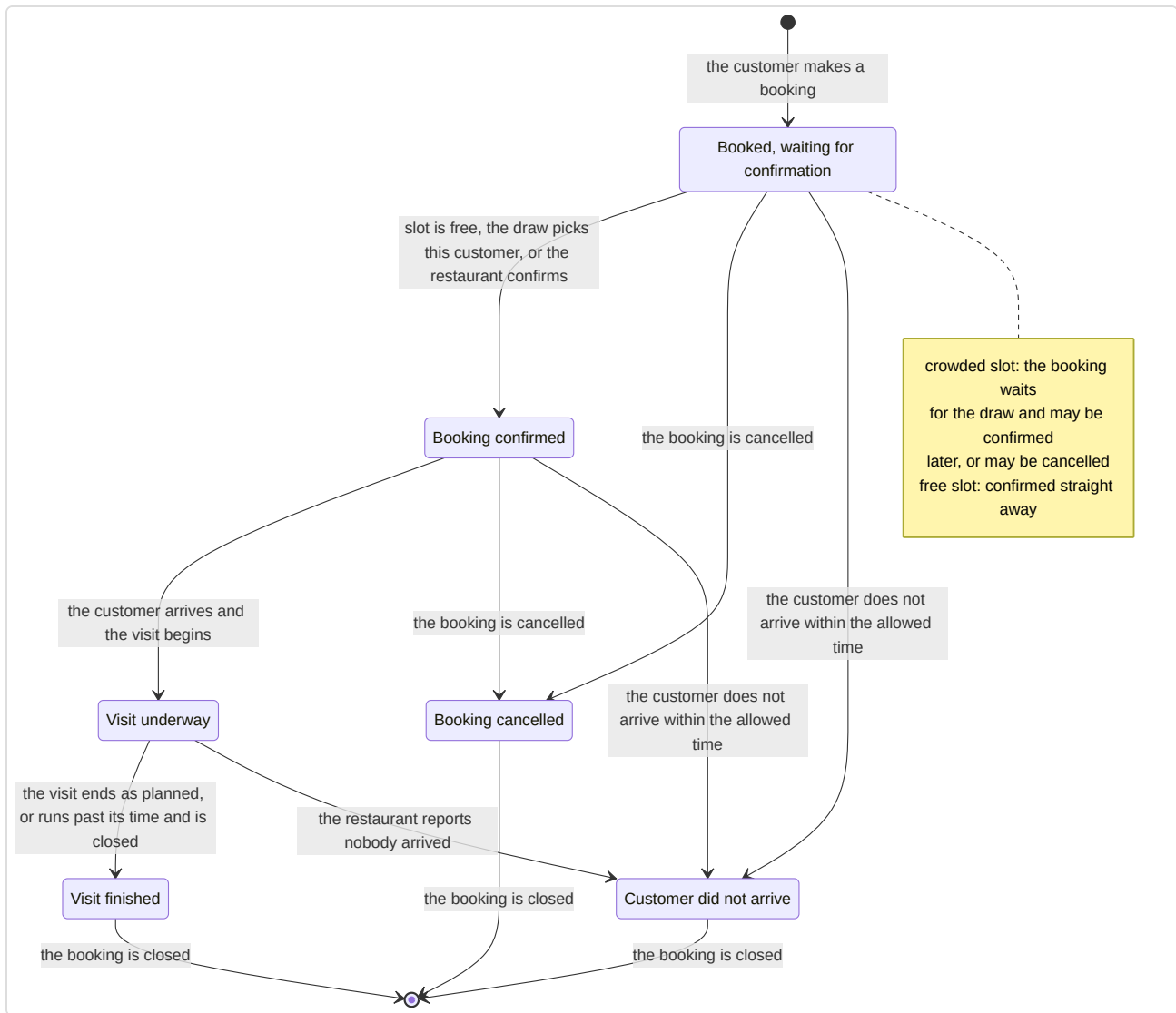




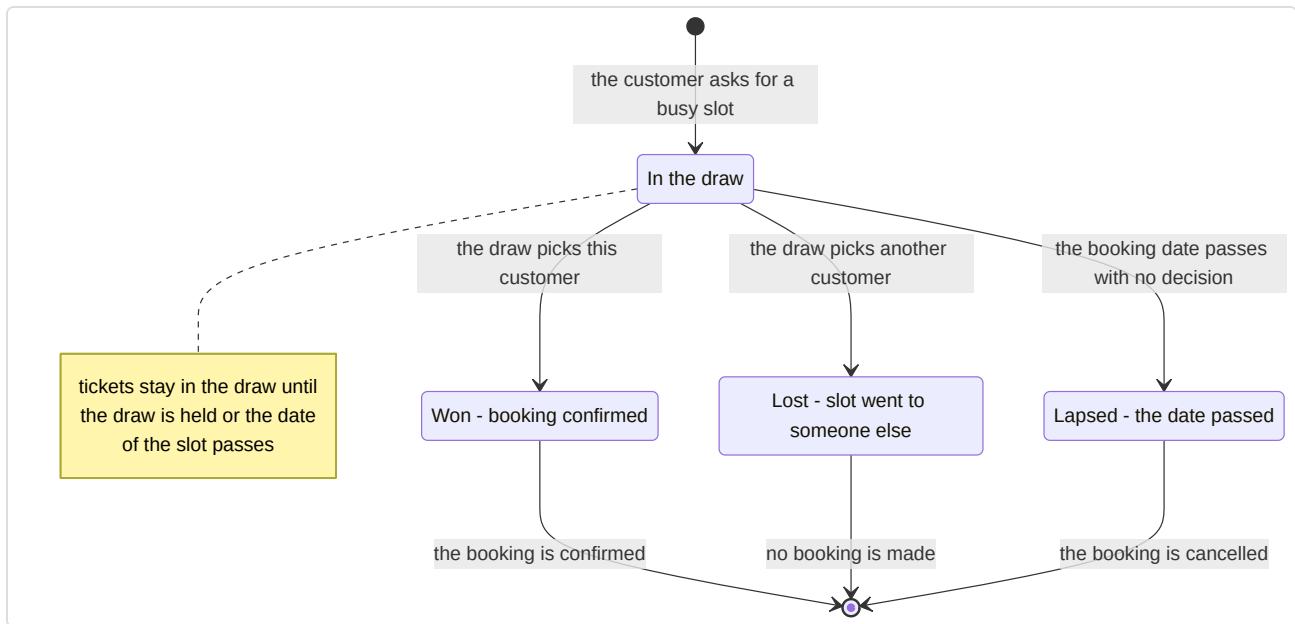
What happened	Cost to the trust score
Customer never arrived	0.70
Customer arrived late	0.40
Customer cancelled late	0.30
Customer overstayed	0.15
Customer usually shows up	the penalty is softened

6. UML State Machine Diagram — What Happens to a Booking

This diagram shows the stages a booking moves through, and what moves it from one stage to the next. It is written for non-technical readers, so everyday words are used instead of internal names.



7. UML State Machine Diagram — A Customer's Ticket in the Draw



8. Process & Trigger Summary

Process	When it happens	Outcome
Booking a table	The customer books a table and time	Confirmed, or waits for the draw
Joining the draw	The customer asks for a busy slot	A ticket is placed in the draw
Holding the draw	Automatically, every minute	One winner, everyone else released
Auto-confirming a free slot	Automatically, after a short wait	Waiting booking confirmed
Recording arrival	The restaurant notes the customer arrived	Visit starts; late arrivals are noted
Reporting a no-show	The restaurant reports the customer never arrived	Customer marked as a no-show
Extending a stay	The customer asks to stay longer	End time moved later
Closing late visits	Automatically, every minute	Overdue visits finished or marked as no-shows
Finishing a visit	The restaurant marks the visit as over	Visit finished
Cancelling	The customer, restaurant or admin cancels	Booking cancelled

Process	When it happens	Outcome
Refreshing trust	After every finished, missed or cancelled visit	Customer's trust profile updated